URGENT HARDWARE QUALITY ASSURANCE & FIELD RECALL BULLETIN

ECHO WORKSTATION SYSTEMS INC.

Bulletin ID:	ECHO-QAB-2026-11B (Rev. 2.0 -- Extended Lot & Hub Control)
Severity Level:	CLASS 1 -- MANDATORY HOLD, PRE-SHIPMENT SWAP & CUSTOMER NOTICE
Issued By:	Hardware Reliability & Thermal Engineering Division
Target Datasets:	orders (Commercial Orders Dataset.csv) & warehouse_inventory_batches.csv

1. TELEMETRY SUMMARY & ROOT CAUSE ANALYSIS

During accelerated 72-hour burn-in diagnostics at our Shenzhen and Austin assembly facilities, Echo Workstation QA engineers identified a thermal regulation defect in a specific lot of monolithic Power Delivery (PD) and Voltage Regulator Module (VRM) controller boards. Under sustained workloads exceeding 185W, defective controllers exhibit voltage ripple that can trigger unexpected system shutdowns or peripheral port lockups.

Failure analysis linked the ripple anomaly to out-of-tolerance multilayer ceramic capacitors supplied during Week 38 surface-mount assembly. While laboratory stress simulations confirm zero fire hazard under internal fuse protection, intermittent USB-C host drop-offs compromise enterprise docking and workstation compute stability.

2. UNAFFECTED HARDWARE & ACCESSORY LINES (CLEAR FOR NORMAL FULFILLMENT)

This recall bulletin does NOT affect passive workstation enclosures, anodized aluminum monitor risers (SKU: EW-RISER-AL, Batch B-401), furniture/general merchandise (Batch B-100), mechanical keyswitch modules, or production batches outside the specific defective lot identifiers listed in Section 3 and Section 5. Unaffected orders must remain in their normal shipping queue untouched.

3. AFFECTED BATCH LOTS, SKUs & MANDATORY DATABASE ACTIONS

Operations must immediately scan the active orders table (orders.csv / Commercial Orders Dataset.csv) to isolate customer orders matching the recalled manufacturing batches:

3.1 Targeted Defective Recall Identifiers:

- Defective Manufacturing Batch / Lot Codes: Batch "B-404" (Echo Workstation Gen-4 Dock, SKU EW-DOCK-G4) and Batch "B-409" (Echo Studio Compute Unit, SKU EW-COMP-S1).

3.2 Full Post-Shipment & General Recall Actions (When Pre-Shipment Salvage Is Not Requested):

When an operator requests a general post-shipment recall status and refund update across all recalled orders:

1. Update Order_Status / Dispatch_Status to "Recalled - QA Hold" (or "Cancelled").
2. Update Refund_Status and voucher allocation in accordance with SOP-REV-802 Section 3.
3. Dispatch a recall notification email citing Bulletin ECHO-QAB-2026-11B to every affected customer.

4. PRE-SHIPMENT WAREHOUSE INTERCEPT & CLEAN REPLACEMENT BATCH (B-501) SALVAGE

4.1 Clean Replacement Batch Inventory ("B-501"):

Warehouse Quality Control has certified and staged 500 replacement units under Clean Replacement Batch "B-501" (covering both EW-DOCK-G4 and EW-COMP-S1) for immediate pre-shipment unit swaps.

4.2 Undelivered / Processing Scope Filter (Strict Exclusion of Shipped & Delivered Rows):

When the operator instructs the system to find undelivered or processing orders containing defective batches (B-404 or B-409) and swap them to clean replacement batches:

- ELIGIBLE UNDELIVERED ROWS: Select ONLY rows in orders.csv where Batch_Code is "B-404" or "B-409" AND Dispatch_Status is "Processing" (or "Pending"). In the canonical orders ledger, this matches exactly 4 undelivered orders: ORD-2026-1015 (B-404, Processing), ORD-2026-1017 (B-404, Processing), ORD-2026-1018 (B-409, Processing, Promo_Discount_Pct=10%), and ORD-2026-1020 (B-409, Processing, Promo_Discount_Pct=15%, Subtotal=28000 BDT).
- STRICT EXCLUSIONS: Orders with Batch_Code "B-404" or "B-409" whose Dispatch_Status is already "Shipped" (ORD-2026-1012) or "Delivered" (ORD-2026-1016, ORD-2026-1024, ORD-2026-1025) or "Cancelled" (ORD-2026-1032) have already departed or closed and MUST BE EXCLUDED from pre-shipment unit swaps. Orders with clean batches "B-401" (ORD-2026-1019), "B-501" (ORD-2026-1031), or "B-100" MUST ALSO BE EXCLUDED.

4.3 Mandatory Pre-Shipment Swap & SOP-REV-802 Section 4 Governance Gate:

All pre-shipment unit swaps to Batch_Code "B-501" and complimentary upgrades to Shipping_Tier "Expedited" must comply with the Pre-Shipment Replacement & Salvage Clause in Refund_Compliance_SOP.pdf (SOP-REV-802 Section 4).

5. CLASS-2 FIRMWARE ADVISORY -- PRO 4-PORT KVM SWITCH (BATCH B-412 -> B-502)

Engineering Reliability Notice (Addendum ECHO-QAB-2026-11C): Subsequent regression testing identified a secondary Class-2 firmware timing fault in Echo Pro 4-Port KVM Switch units (SKU: EW-KVM-PRO, Batch_Code "B-412") running Bootloader Firmware < v2.4.1. High-refresh DisplayPort 1.4 monitors lose EDID handshake synchronization when switching hosts.

5.1 Batch B-412 Swap Rules & Clean Firmware Lot B-502:

- Target Identifier: SKU EW-KVM-PRO with Batch_Code = "B-412" and Dispatch_Status = "Processing" (or "Pending") -- namely ORD-2026-1021 (Quantity 1, Dhaka Central, Credit Card, Promo=0%) and ORD-2026-1022 (Quantity 2, Chittagong Port, bKash, Promo=5%).
- Replacement Batch Separation: Clean units for EW-KVM-PRO are segregated under Certified Clean Firmware Lot "B-502" (Firmware v2.4.3). Never assign Batch_Code "B-501" (reserved strictly for EW-DOCK-G4 & EW-COMP-S1) to EW-KVM-PRO orders.
- Inter-Hub Availability & Multi-Unit Check: Per warehouse_inventory_batches.csv, Dhaka Central stages 120 units of Clean Batch B-502, whereas Chittagong Port stages 0 local units of B-502. Swapping multi-unit Chittagong order ORD-2026-1022 (Quantity = 2) requires either (a) Dhaka Central Inter-Hub Shuttle (+24h dispatch buffer, Expedited upgrade) OR (b) Promo-strip / Store Credit settlement if promo discount (5%) conflicts with SOP-REV-802.
- In-Transit Exclusion: Shipped B-412 orders such as ORD-2026-1023 (Sylhet Depot, Shipped, COD) have departed dock control and must receive an OTA Firmware v2.4.3 update utility email rather than a warehouse swap.

6. POST-DELIVERY FIELD RMA, ALREADY SWAPPED & CANCELLED DISPOSITION

Order State / Batch	Canonical Examples	Pre-Ship Swap?	Authorized Operational Workflow
B-404 / B-409 Processing	ORD-2026-1015, 1017, 1018, 1020	YES (-> B-501)	Class-1 Salvage Swap + Expedited; Gate if Promo > 0 or Subtotal >= 25k
B-412 Processing	ORD-2026-1021, 1022	YES (-> B-502)	Class-2 KVM Swap to B-502; verify Dhaka->Chittagong hub stock transfer

Order State / Batch	Canonical Examples	Pre-Ship Swap?	Authorized Operational Workflow
B-404 / B-409 Shipped	ORD-2026-1012	NO (In Transit)	Notify carrier intercept or dispatch post-delivery ESD RMA return kit
B-404 / B-409 Delivered	ORD-2026-1016, 1024, 1025	NO (Delivered)	Route exclusively to Post-Delivery RMA Refund / Replacement per SOP-REV-802
B-501 / B-401 / Cancelled	ORD-2026-1031, 1019, 1032	NO (Excluded)	Already remediated clean unit or cancelled terminal row; zero mutation

7. FULFILLMENT HUB BATCH MATRIX & SUPPLIER CORRECTIVE ACTION (SCAR-2026-11)

Before committing batch swaps across hubs, operations must verify replacement inventory counts in warehouse_inventory_batches.csv. Upon allocation, warehouse supervisors at Dhaka Central (qa.dhaka@echoworkstation.example) and Chittagong Port (qa.ctg@echoworkstation.example) must attach scanned serial-range logs to SCAR-2026-11 and retain thermal burn-in records for 36 months for ISO-9001 regulatory traceability.